

SAI SRUTHI CHALASANI

Software Engineer • Backend & Distributed Systems • Applied AI • Cloud & DevOps

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SUMMARY

Software Engineer with **5+ years** designing, building, and operating production backend and **applied AI** systems across logistics, financial services, and automotive. Builds in **Python** and **Java** — distributed **microservices** with **Spring Boot**, **REST APIs**, and **Apache Kafka**, alongside **LLM-powered applications** using **RAG**, **vector search**, **MCP**, and **LangGraph** agentic workflows. Owns the **full SDLC**: requirements and API design, containerized delivery on **Docker** and **Kubernetes**, **CI/CD** with **Jenkins** and **GitHub Actions**, and production **observability** with **Dynatrace** and **ELK** on **AWS** and **GCP**. **AWS Certified Solutions Architect** with a record of replacing manual enterprise workflows with automation, cutting manual effort **40-70%**.

PROFESSIONAL EXPERIENCE

UPS — United Parcel Service | *Senior Software Development Engineer*

Parsippany, NJ | Jul 2023 – Present

- Built a **production GenAI knowledge assistant** in **Python** — **RAG** pipeline with **hybrid BM25 + vector retrieval** over **pgvector**, semantic chunking, **embedding generation**, and **re-ranking**, behind a **model-agnostic LLM orchestration** layer over **Claude** and **OpenAI** APIs, integrating **IBM ODM**, **Azure DevOps**, and engineering documentation via **MCP**, collapsing three-system lookups into a single **grounded, citation-backed** query.
- Engineered an **agentic code-review and remediation workflow** with **stateful agent orchestration** in **LangGraph**: **SonarQube** findings drive **multi-step tool-calling** and **structured outputs** against the **Claude API**, agent-authored fixes run **automated test execution** and **self-validation gates** in **GitHub Actions**, and changes ship under **human-in-the-loop** approval — cutting review effort **~40%**.
- Implemented **AI security and governance controls** across **GenAI** and **agentic workflows** using **Guardrails AI**, aligning with **NIST AI RMF** and **OWASP LLM** security guidance through **prompt-injection defenses**, **input/output validation**, **PII and sensitive-data safeguards**, **least-privilege tool access**, **automated evaluation gates**, **auditability**, and **human-in-the-loop approvals** for high-impact agent actions.
- Automated the annual **General Rate Increase** across **~10M pricing records**, building extraction, transformation, and validation pipelines for **large-scale** rule processing over **IBM ODM** decision tables with **rule-lifecycle versioning** and release automation through the **Decision Center Java client API** — cutting manual effort **70%** and deployment errors **50%**.
- Architected the **ODM Deployment Scheduler**, a **release-orchestration** service performing **delta detection** across completed **Decision Center** change activities, **automated test generation** from detected changes, **regression execution** behind **CI/CD-style quality gates**, and **artifact publishing** of **RuleApp JARs** to **GCP Cloud Storage** and the **Rule Execution Server (RES)**.
- Scaled **cloud-native, high-throughput microservices** and **REST APIs** for global package rating, billing, and quoting on **Java** and **Spring Boot**, with **asynchronous event streaming** through **Apache Kafka** and **React/Angular** front-end integrations, reducing API latency **35%** and improving revenue tracking accuracy **30%**.

JPMorgan Chase & Co. | *Software Development Engineer*

New York, NY | Jan 2023 – May 2023

- Modernized the Credit Card Subscription and Notification Platform through **AWS cloud migration**, decomposing legacy workloads into independently deployable, **horizontally scalable Spring Boot microservices** with **React** front-end components, sustaining **millions of daily transactional alerts**.
- Optimized backend **data-access and API integration** paths for high-volume financial workloads — schema and query tuning across **PostgreSQL**, **SQL** workload profiling, and **REST API** integration — improving query performance by **35%**.
- Built **automated build, test, and deployment pipelines** with **API contract validation** through **Postman** and **Swagger/OpenAPI**, enforcing **release quality gates** and improving **deployment reliability** across independently deployable **microservices**.
- Diagnosed production **latency and reliability** issues across high-volume **alert-delivery paths** using **Dynatrace distributed tracing**, tuning throughput across distributed services in **Agile/Scrum** delivery with cloud architecture and platform teams.

Hyundai Mobis | *Software Developer*

Hyderabad, India | Jan 2020 – Dec 2021

- Developed **Remote Smart Parking Assist (RSPA)** embedded software using **Model-Based Design** in **MATLAB/Simulink**, implementing, debugging, and validating **Parking Slot Detection** and **Parking Path Control algorithms** on production vehicle platforms.
- Authored **Parking Slot Detection** logic shipped in the **GENESIS G90 ECU**, clearing functional, unit, and **model-in-the-loop (MIL)** validation with **Polyspace** static analysis and **CARSIM** vehicle simulation against automotive **functional-safety** standards.
- Built automated **test harnesses** and **regression automation** for embedded parking-assist modules, scripting simulation, **data processing**, and validation workflows in **Python** and **MATLAB**, executed in **Agile/Scrum** sprints under **Git** version control and shortening algorithm validation cycles.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript, TypeScript, Bash/Shell, MATLAB

Backend & Distributed Systems: Spring Boot, REST APIs, microservices, event-driven architecture, Apache Kafka, event streaming, ETL and data pipelines, batch processing, data validation

AI & Agentic Systems: Generative AI, Agentic AI, LLM application development, Retrieval-Augmented Generation (RAG), hybrid retrieval (BM25 + vector), vector search, pgvector, embeddings, semantic chunking, re-ranking, context and prompt engineering, Model Context Protocol (MCP), LangGraph, agent and tool orchestration, tool/function calling, structured outputs, human-in-the-loop workflows, agent evaluation and self-validation loops, AI governance, AI security, Responsible AI, Guardrails AI, NIST AI RMF, OWASP LLM Security, prompt-injection defense, Claude API, OpenAI API

Databases: PostgreSQL, Oracle, MySQL, MongoDB

Cloud, Platform & DevOps: AWS (EC2, S3, RDS, Lambda, IAM, CloudWatch), GCP (Cloud Storage), Kubernetes, OpenShift, Docker, Terraform, Jenkins, GitHub Actions, Azure DevOps, CI/CD, AI application deployment, IBM ODM, Decision Center, Rule Execution Server (RES), Apache POI, Jsoup

Testing, Observability & Quality: Dynatrace, Grafana, ELK Stack, SonarQube, JUnit 5, Mockito, Postman, Swagger, JMeter, Model-Based Design (MATLAB/Simulink), Polyspace, CARSIM, Agile/Scrum

EDUCATION

Stevens Institute of Technology — M.S., Computer Science

Hoboken, NJ | Jan 2022 – May 2023

Amrita Vishwa Vidyapeetham — B.Tech, Electronics & Instrumentation

Coimbatore, India | Jul 2016 – Apr 2020